

Percentage, Profit and Loss, Interest

By Pranav Thorat

Percentage

- **Ratio** relation W.R.T. **Hundred**
- If $A:B=X:100$ then A is $X\%$ of B

Simplification

- $45\% \text{ of } 840 + 35\% \text{ of } 620 = ?$

- $75\% \text{ of } 920 - 55\% \text{ of } 640 + 25\% \text{ of } 360 = ?$

- $68\% \text{ of } 750 - 42\% \text{ of } 450 = ?$

- $85\% \text{ of } 1400 - 65\% \text{ of } 820 = ?$

- $24\% \text{ of } 1350 + 18\% \text{ of } 650 = ?$

- $32\% \text{ of } 450 + 54\% \text{ of } 350 = ?$

Simplification

- $15\% \text{ of } 1800 + ? \% \text{ of } 600 = 420$
- $92\% \text{ of } 550 - 38\% \text{ of } 450 = ?$
- $56\% \text{ of } 950 - 28\% \text{ of } 450 = ?$
- $12\% \text{ of } 850 + 26\% \text{ of } 450 - 14\% \text{ of } ? = ?$
- $48\% \text{ of } 650 + 72\% \text{ of } 250 = ?$
- $60\% \text{ of } 40\% \text{ of } 8500 = ?$

- In a town, females constitute 40% of the total population and are 30,000 fewer in number than males. Based on this information, answer the following questions:
 - a) Find the total population of the town.
 - b) Find the male population of the town.
 - c) If 55% of the females are married, find the number of unmarried females in the town.
 - d) Find the number of married females and the ratio of married females to the total male population.

Find the total population of the town

$$F = 40\% \text{ of } T$$

$$M = 60\% \text{ of } T$$

$$M - F = 30000$$

$$20\% \text{ of } T = 30000$$

$$\frac{20}{100} \times T = 30000$$

$$T = 150000$$

Direct Variable calculation

Find the male population of the town.

$$20\% \text{ of } T = 30000$$

$$60\% \text{ of } T = ?$$

$$\frac{60\% \text{ of } T}{20\% \text{ of } T} \times 30000 = L$$

Cross Multiplication

If 55% of the females are married, find the number of unmarried females in the town.

$$MF = 55\% \text{ of } F$$

$$UF = 45\% \text{ of } F$$

$$F = 40\% \text{ of } T$$

$$UF = 45\% \text{ of } 40\% \text{ of } T$$

$$\frac{45}{100} \times \frac{40}{100} \times 150000 = ?$$

$$UF = 27000$$

Chain percentage equation

Find the number of unmarried males and the ratio of married females to the total male population.

Basic Common sense

$$MM = MF$$

$$UM = M - MM$$

$$UM = 60\% \text{ of } T - 55\% \text{ of } 40\% \text{ of } T$$

$$UM = 57000$$

$$MF:M = \frac{55\% \text{ of } 40\% \text{ of } T}{60\% \text{ of } T}$$

$$MF:M = 11:30$$

Statement to equations

1. In a school if the number of girls are 37% of the total and are 1352 less than the boys. Then find out the total number of students of their school as well as boys?
2. In an election the candidate who got 62% of the total votes defeated his only rival by 2304 votes. Then find out the votes received by the loser?
3. In an election the candidate who got 56% of the valid votes defeated his only rival by 1440 votes. Then find out total number of votes of the election if 20% of them were invalid?

In a school if the number of girls are 37% of the total and are 1352 less than the boys. Then find out the **total number** of students of their school as well as **number of boys**?

$$G = 37\% \text{ of } T$$

$$B = T - 37\% \text{ of } T = 63\% \text{ of } T$$

$$B - G = 1352$$

$$63\% \text{ of } T - 37\% \text{ of } T = 1352$$

$$26\% \text{ of } T = 1352$$

$$\frac{26}{100} \times T = 1352$$

$$26\% \text{ of } T = 1352$$

$$63\% \text{ of } T = B$$

In an election the candidate who got 62% of the total votes defeated his only rival by 2304 votes. Then find out the votes received by the loser?

$$W = 62 \% \text{ of } T$$

$$L = 38 \% \text{ of } T$$

$$W - L = 2304$$

$$24 \% \text{ of } T = 2304$$

$$38 \% \text{ of } T = L$$

$$\frac{38 \% \text{ of } T}{24 \% \text{ of } T} \times 2304 = L$$

1. In an election the candidate who got 56% of the valid votes defeated his only rival by 1440 votes. Then find out total number of votes of the election if 20% of them were invalid?

$$W = 56 \% \text{ of } V$$

$$L = 44 \% \text{ of } V$$

$$12 \% \text{ of } V = 1440$$

$$20 \% \text{ of } T = \text{In } V$$

$$V = 80 \% \text{ of } T$$

$$12 \% \text{ of } 80 \% \text{ of } T = 1440$$

$$\frac{12}{100} \times \frac{80}{100} \times T = 1440$$

Value Changes Q's



Value change problems

Initial Value $\xrightarrow{P\% \uparrow}$ Final Value

- **Introductory question-**

If last year population of a town was **5000** and it was **increased** by **10%** throughout the year then find out the population today?

Today's Popⁿ = Last yr Popⁿ + Changed Popⁿ

Today's Popⁿ = 5000 + **10 %** of 5000

= (100 + **10**) % of 5000 OR 5000 $(1 + \frac{10}{100})$

Replace **Increased** by **Decreased**

$\frac{+10}{100}$ will be replaced by $\frac{-10}{100}$

General formula

- $V_f = (100 + P)\% \text{ of } V_i = V_i \left(1 + \frac{P}{100}\right)$
- V_f :- value after percentage change
- V_i :- value before percentage change
- P :- percentage change given in the question
- Take 'P' positive- increase, more, great, **Profit**, tax, mark-up
- Take 'P' negative- decrease, less, small, **Loss**, discount, mark-down, Spent

- The population is increased by 20% in 2nd year year?



$$X = V_i \left(1 + \frac{P}{100}\right)$$

$$V_f = X \left(1 + \frac{Q}{100}\right)$$

$$V_f = V_i \left(1 + \frac{P}{100}\right) \left(1 + \frac{Q}{100}\right)$$

$$P\% \uparrow \rightarrow Q\% \uparrow \Rightarrow Np\% \uparrow$$

$$N_p = P + Q + \frac{PQ}{100}$$

$$V_f = (100 + N_p) \times V_i \text{ OR}$$

$$= V_i \left(1 + \frac{N_p}{100}\right)$$

Successive percentage value change

- If a value is provided with more than one percentages change one after another in a question.
- Ex- population of a town increased by P% in first year and by Q% in second year
- General formula-: $V_f = V_i \left(1 + \frac{P}{100}\right) \left(1 + \frac{Q}{100}\right)$
- Net percentage(Np) formula-: $Np = P + Q + \frac{PQ}{100}$
- WHEN A VALUE [Initial(V_i) OR Final (V_f)] HAS BEEN ASKED THEN USE **GENERAL FORMULA** AND WHEN ONLY PERCENTAGE CHANGE HAS BEEN ASKED THEN USE **NP FORMULA**

Step Change(Np=0)

- When two successive percentage changes gives **Np=0** on the initial value then those two percentage changes are called as 'Step Changes' of 'each other'.

- $P\% \uparrow \leftrightarrow \frac{100P}{100+P}\% \downarrow$

- $Q\% \downarrow \leftrightarrow \frac{100Q}{100-Q}\% \uparrow$

Successive Practise

1. $20\%\uparrow \rightarrow 10\%\uparrow \Rightarrow x\%\uparrow$
2. $20\%\uparrow \rightarrow 30\%\downarrow \Rightarrow x\%\downarrow$
3. $20\%\downarrow \rightarrow 25\%\downarrow \Rightarrow x\%\downarrow$
4. $30\%\uparrow \rightarrow 20\%\downarrow \Rightarrow x\%\uparrow$
5. $20\%\uparrow \rightarrow x\%\downarrow \Rightarrow 8\%\uparrow$
6. $20\%\uparrow \rightarrow 25\%\uparrow \rightarrow 10\%\downarrow \Rightarrow x\%\uparrow$
7. $20\%\uparrow \rightarrow x\%\downarrow \rightarrow 25\%\uparrow \Rightarrow 35\%\uparrow$

Successive Practise

- $20 + 10 + (20 \times 10)/100 = 30 + 2 = 32\% \uparrow$
- $20 - 30 + (20 \times -30)/100 = -10 - 6 = 16\% \downarrow$
- $-20 - 25 + (-20 \times -25)/100 = -45 + 5 = 40\% \downarrow$
- $30 - 20 + (30 \times -20)/100 = 10 - 6 = 4\% \uparrow$
- $20 - x - (20x)/100 = 8 \Rightarrow 12 = 1.2x \Rightarrow x = 10\%$
- $20 + 25 + (20 \times 25)/100 = 50\% \uparrow$; then $50 - 10 + (50 \times -10)/100 = 40 - 5 = 35\% \uparrow$
- $(1.20) \times (1 - x/100) \times (1.25) = 1.35 \Rightarrow 1.50 \times (1 - x/100) = 1.35 \Rightarrow 1 - x/100 = 0.90 \Rightarrow x = 10\%$

Step Change

1. 25%↑

$$1. (100 \times 25) / (100 + 25) = 20\% \downarrow$$

2. 20%↑

$$2. (100 \times 20) / (100 + 20) = 50 / 3 = 16.66\% \downarrow (16\frac{2}{3}\% \downarrow)$$

3. 40%↓

$$3. (100 \times 40) / (100 - 40) = 66.66\% \uparrow (66\frac{2}{3}\% \uparrow)$$

4. 25%↓

$$4. (100 \times 25) / (100 - 25) = 100 / 3 = 33.33\% \uparrow (33\frac{1}{3}\% \uparrow)$$

5. 125%↑

$$5. (100 \times 125) / (100 + 125) = 500 / 9 = 55.55\% \downarrow$$

6. 50%

$$6. (100 \times 50) / (100 + 50) = 100 / 3 = 33.33\% \downarrow$$

1. After increasing the population by 25% of a town throughout the year it became 62500 then find out how much it was one year ago?
2. After the tax of 35% a product was sold at rupees 729 then find out its MRP?
3. If 1089 is 65% more than X, then find x?
4. Ramesh spent 12% of his salary on room rent if he is left with rupees 19360 then how much is his room rent?
5. Population of a town increased by 20% in first year and by 10% in second year if it was 26400 after those two years then how much it was 2 years ago?
6. After the tax of 10% A shopkeeper offered a discount of 20% on a product if he sold that product at rupees 3872 then find out the MRP of the product?
7. Rohit spend 10% of his salary on room rent 20% of the remaining on his daily shopping and 25% of the remaining he invested in a scheme, if he is left with rupees 21600 in his account then how much was his salary and also how much amount he has invested in that scheme?

After increasing the population by 25% of a town throughout the year it became 62500 then find out how much it was one year ago?

$$\bullet V_i \xrightarrow{P\% \uparrow} V_f$$

$$\bullet V_i \xrightarrow{25\% \uparrow} 62500$$

$$\bullet 62500 = (100 + 25)\% \text{ of } V_i$$

$$\bullet 62500 = \frac{125}{100} \times V_i$$

After the tax of 35% a product was sold at rupees 729 then find out its MRP?

• $V_i \xrightarrow{P\% \uparrow} V_f$

• $V_i \xrightarrow{35\% \uparrow} 729$

• $729 = (100 + 35)\% \text{ of } V_i$

• $729 = \frac{135}{100} \times V_i$

1. If 1089 is 65% more than X, then find x?

1. Ramesh spent 12% of his salary on room rent if he is left with rupees 19360 then how much is his room rent?

$$\begin{aligned}88\% \text{ of } S &= 19360 \\12\% \text{ of } S &=?\end{aligned}$$

$$\frac{12}{88} \times 19360$$

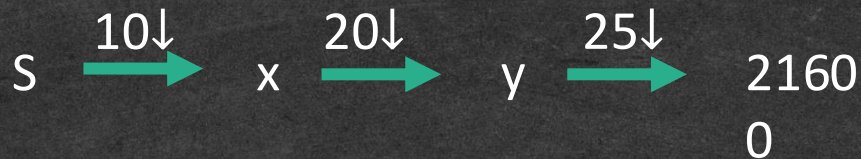
1. Population of a town increased by 20% in first year and by 10% in second year if it was 26400 after those two years then how much it was 2 years ago?

$$26400 = \frac{120}{100} \times \frac{110}{100} \times V_i$$

1. After the tax of 10% A shopkeeper offered a discount of 20% on a product if he sold that product at rupees 3872 then find out the MRP of the product?

$$3872 = \frac{110}{100} \times \frac{80}{100} \times V_i$$

- Rohit spend 10% of his salary on room rent 20% of the remaining on his daily shopping and 25% of the remaining he invested in a scheme, if he is left with rupees 21600 in his account then how much was his salary and also how much amount he has invested in that scheme?



- $$s \times \frac{90}{100} \times \frac{80}{100} \times \frac{75}{100} = 21600$$

- For investment

$$75\% \text{ of } y = 21600$$

$$25\% \text{ of } y = ?$$

Step change question where solution is same but Language defers

1. If a is 25% more than B ; Then b is what percent less than a ?
2. Price of a product is increased by 25% by the shopkeeper but due to the drop in the sale he decided to bring it back to its original value then by what percent now he will have to reduce the price of the product?
3. Price of a commodity is increased by 25% then by what percent its consumption should be decreased so that the total expenditure on that product doesn't get affected
4. A metallic rectangular plate is melted and converted into another rectangular shape without wasting or adding any metal. If the new rectangular plate has length 25% more than the previous one then by what percent its width might have reduced
5. To nullify the tax of 25% how much discount A shopkeeper must offer

1. If A is 33.33 percent less than B then b is what percent greater than a
2. A shopkeeper offer a discount of 33.33 percent on all the products then by what percent his sale should increase so that his income does not get affected
3. X varies inversely with Y;



Buy: Price @ which the product is Bought OR Amount Spent/Given: **Cost Price**

Sell: Price @ which the product is Sold OR Amount Received/Earned: **Selling Price**

Profit

$$SP > CP$$

$$SP - CP > 0$$

$$SP - CP \text{ is } + \text{ve}$$

Loss

$$SP < CP$$

$$SP - CP < 0$$

$$SP - CP \text{ is } - \text{ve}$$

Relation with Value change

$$V_f = (100 + P)\% \text{ of } V_i = V_i \left(1 + \frac{P}{100}\right)$$

$$SP = (100 + P)\% \text{ of } CP \text{ OR } CP \left(1 + \frac{P}{100}\right)$$

1. After selling an article for rs 350 A shopkeeper made the profit of 25% then find out the cost price of that article
2. after selling an article for rupees 729 A shopkeeper made the profit of 35% then find out at what price he should have sold that article to have the profit of 50%
3. A shopkeeper sold an article for some cost and bear the loss of 9% had he sold it for rupees 225 more he would have had the profit of 6%. then find out for how much price he should sell it to have the profit of 30%?
4. A shopkeeper markup the price of a product by 400% and offer the discount of 50% then find out the percentage profit he was having
5. Atul sold a chair to Bablu with the profit of 20% Bablu sold same chair to Chandu with the profit of 35%. Chandu has given rupees 648 to Bablu for the same chair then find out the amount of profit Bablu made from his transaction?

1. After selling an article for rs 350 A shopkeeper made the profit of 25% then find out the cost price of that article

$$Sp = Cp \left(1 + \frac{P}{100}\right)$$

$$350 = C_P \times \frac{125}{100}$$

1. after selling an article for rupees 729 A shopkeeper made the profit of 35% then find out at what price he should have sold that article to have the profit of 50%

$$135\% \text{ of } C_P = 729$$

$$150\% \text{ of } C_P = ?$$

$$\frac{150}{135} \times 729$$

1. A shopkeeper sold an article for some cost and bear the loss of 9% had he sold it for rupees 225 more he would have had the profit of 6%. then find out for how much price he should sell it to have the profit of 30%?

$$X = 91\% \text{ of } C_p$$

$$X + 225 = 106\% \text{ of } C_p$$

$$\text{Eqn 2} - \text{Eqn 1}$$

$$15\% \text{ of } C_p = 225$$

$$130\% \text{ of } C_p = ?$$

1. A shopkeeper markup the price of a product by 400% and offer the discount of 50% then find out the percentage profit he was having

$$400 + (-50) + \frac{(400) \times (-50)}{100}$$

- Atul sold a chair to Bablu with the profit of 20%
Bablu sold same chair to Chandu with the profit of 35%. Chandu has given rupees 648 to Bablu for the same chair then find out the amount of profit Bablu made from his transaction?

$$135\% \text{ of } B = 648$$

$$35\% \text{ of } B = ?$$

No of articles Question

1. If selling price of 15 notebooks is equal to cost price of 20 notebooks then find out the percentage profit of that bargain
2. shopkeeper bought 5 lemons for 4 rupees and sold 4 of them for 5 rupees, then find out the percentage profit he was having
3. After selling 50 bicycles A shopkeeper gains the selling price of 10 bicycles then find out the percentage profit he had. If another shopkeeper gain the cost price of 10 bicycles then find out the percentage profit he had

1. If selling price of 15 notebooks is equal to cost price of 20 notebooks then find out the percentage profit of that bargain

$$15 S_p = 20 C_p$$

$$S_p = \frac{4}{3}C_p = C_p\left(1 + \frac{P}{100}\right)$$

$$\frac{4}{3} = 1 + \frac{P}{100}$$

$$P = 33.33\% \uparrow$$

1. shopkeeper bought 5 lemons for 4 rupees and sold 4 of them for 5 rupees, then find out the percentage profit he was having

Bought -: 5 lemons for 4 Rs-: 1 lemon price = $\frac{4}{5}$

Sold -: 4 lemons for 5 Rs-: 1 lemon price = $\frac{5}{4}$

$$\frac{5}{4} = \frac{4}{5} \left(1 + \frac{P}{100} \right)$$
$$\frac{25}{16} = 1 + \frac{P}{100}$$

$$P = 56.25 \% \text{ Profit}$$

1. After selling 50 bicycles A shopkeeper gains the selling price of 10 bicycles then find out the percentage profit he had. If another shopkeeper gain the cost price of 10 bicycles then find out the percentage profit he had

$$50 S_p = 50 C_p + 10 S_p$$

$$40 S_p = 50 C_p$$

$$S_p = \frac{5}{4} C_p = C_p \left(1 + \frac{P}{100} \right)$$

$$P = 25 \% \text{ Profit}$$

Allegation of Profit and Loss

1. shopkeeper bought a bullock and a cart for 60000 if he sold Bullock for the profit of 11 % and cart with the profit of 41% if overall he got the profit of 23% then find out the cost price of the cart
2. In what ratio A shopkeeper should mix the two types of sugar costing rupees 32 per kg and rupees 48 per kg e so that if he sells the mixture at rupees 45 per kg he will have the profit of 25%
3. A retailer buys 260 eggs. He sells some of them at a profit of 20% and the remaining at a profit of 30%. If he gains 24% then how many eggs did he sell at the profit of 30%?

1. shopkeeper bought a bullock and a cart for 60000 if he sold Bullock for the profit of 11 % and cart with the profit of 41% if overall he got the profit of 23% then find out the cost price of the cart

1. In what ratio A shopkeeper should mix the two types of sugar costing rupees 32 per kg and rupees 48 per kg e so that if we sell the mixture at rupees 45 per kg he will have the profit of 25%

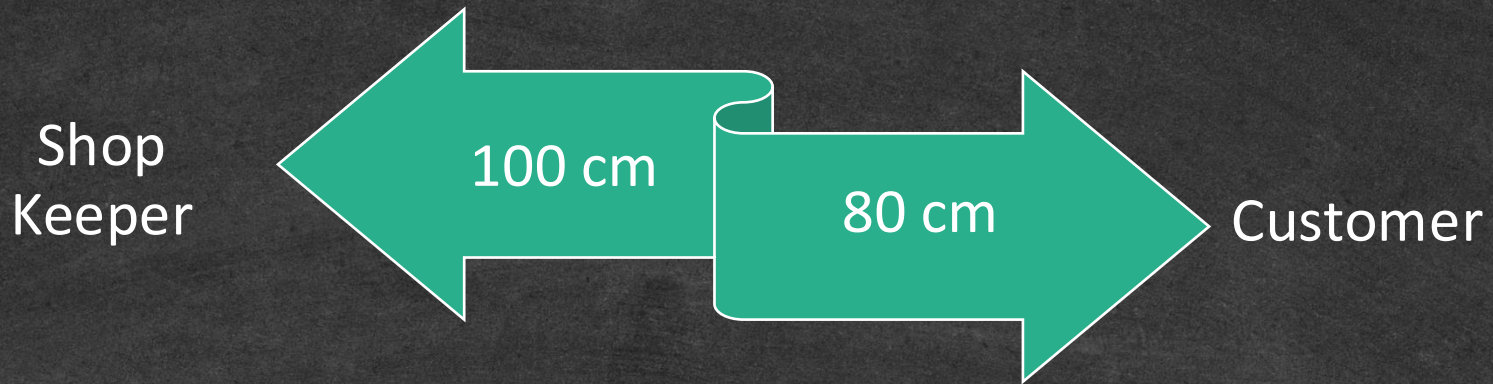
1. A retailer buys 260 eggs. He sells some of them at a profit of 20% and the remaining at a profit of 30%. If he gains 24% then how many eggs did he sell at the profit of 30%?

1. In a stable some hens and buffalos are kept. If total number of heads in that stable is 200 and total number of legs are 550. Then find out number of hens in that stable?

Cheating

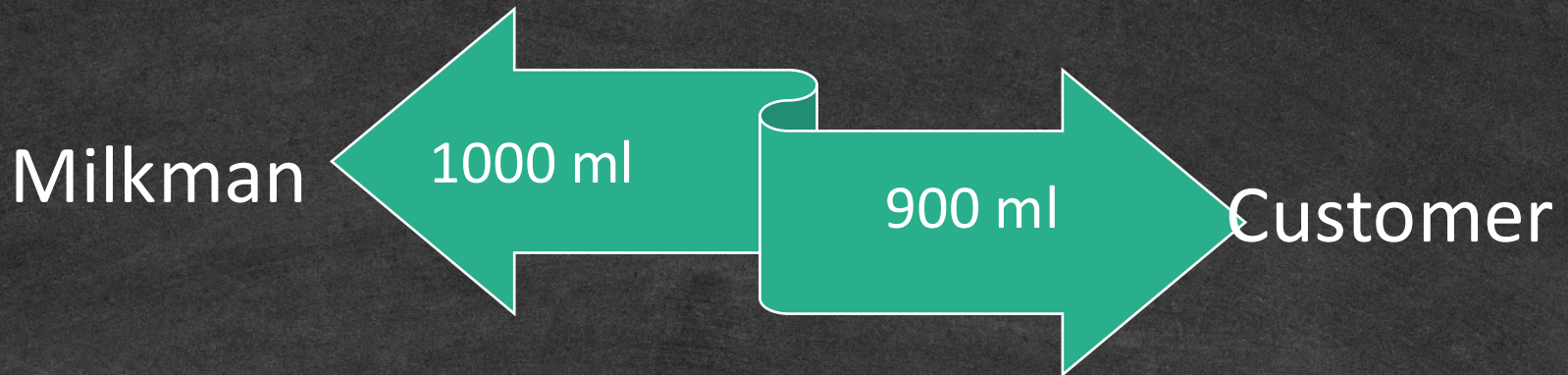
1. To increase the sale a cloth vendor told his customers that he will be selling all the clothing products at its cost price but to ensure some profit he used a faulty metre rod of 80 CM then find percentage of profit he was having
2. A dishonest milkman cheated his customers by using a 900 ml of flask and told them that it is a flask of 1 litre then find out the percentage of profit assured by doing so.
3. A shopkeeper had a faulty weighing machine which used to show 900 grams instead of 1 kg then find out the percentage of profit or loss it was causing on his overall business

1. To increase the sale a cloth vendor told his customers that he will be selling all the clothing products at its cost price but to ensure some profit he used a faulty metre rod of 80 CM then find percentage of profit he was having



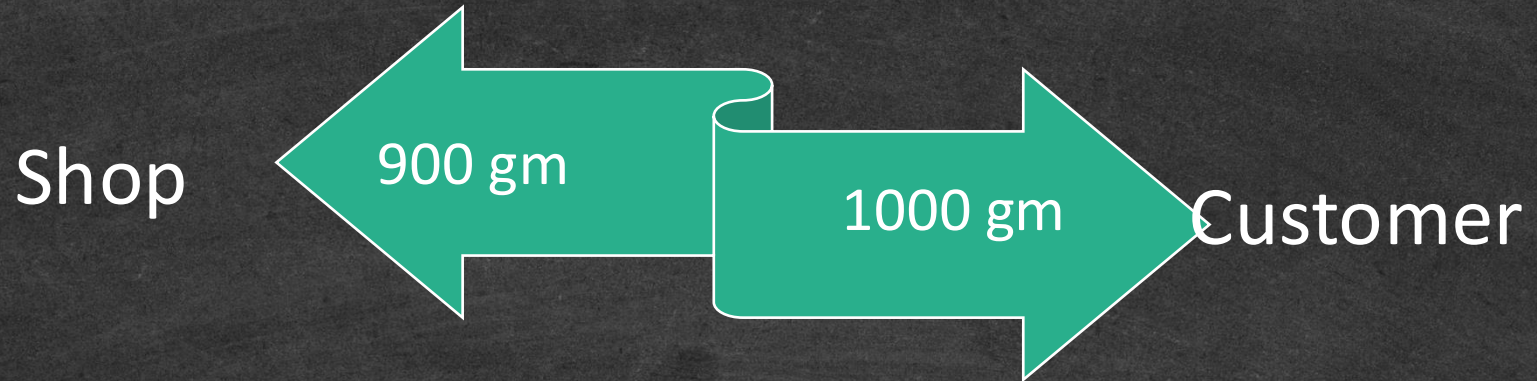
$$100\text{cm} = 80\text{cm} \left(1 + \frac{P}{100} \right)$$

1. A dishonest milkman cheated his customers by using a 900 ml of flask and told them that it is a flask of 1 litre then find out the percentage of profit assured by doing so.



$$1000 \text{ ml} = 900 \text{ ml} \left(1 + \frac{P}{100} \right)$$

1. A shopkeeper had a faulty weighing machine which used to show 900 grams instead of 1 kg then find out the percentage of profit or loss it was causing on his overall business



$$900 \text{ gm} = 1000 \text{ gm} \left(1 + \frac{P}{100} \right)$$